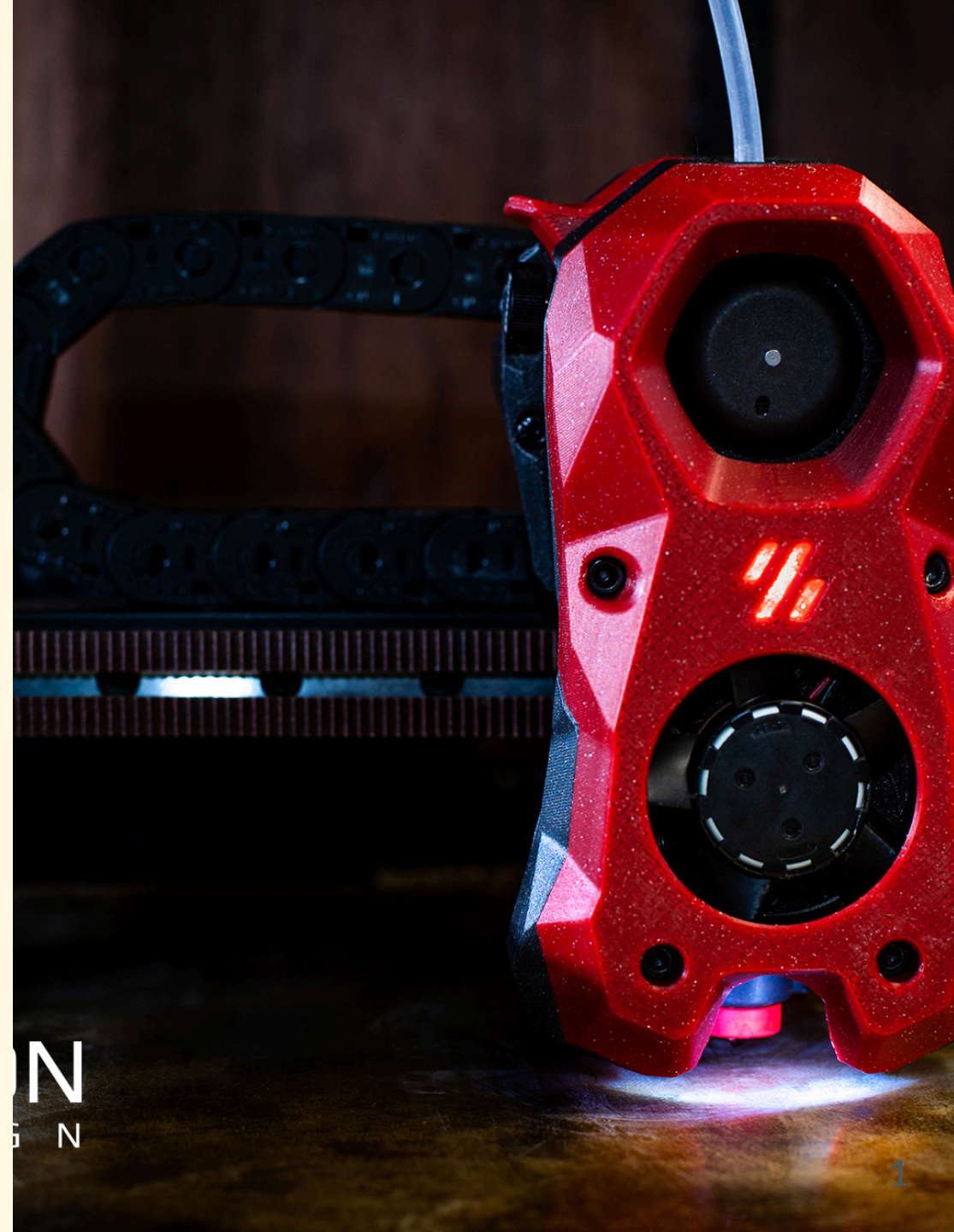


Building a Voron 2.4

A precision, multi-material 3D
printer — the plan, the options, and
the costs

A hobby build proposal • self-sourced • done in phases



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Executive Summary

I'd like to build a new printer — a self-sourced, enclosed, precision printer.

What	A 300 mm Voron 2.4 R2 (0.4 mm, multi-material, enclosed)
Why	The fun of building + multimaterial printing
Cost	≈ \$1,850 core, ≈ \$2,040 after multi-material
How	In phases — always a working printer

Prices are working estimates, being finalized part-by-part.

Why do this?

- **The build itself is the point.** Getting the Kobra Max running was a blast — a Voron is a bigger, more satisfying version of that.
- **It fills a gap.** Our printers are single-material and fussy. A Voron is:
 - **Multi-material** — multi-color / multi-part prints in one go
 - **Enclosed** — can print ABS/ASA (stronger, heat-resistant parts)
 - **Precise & fast** — CoreXY motion, cleaner than our current machines
- **It's a learning project** the whole household benefits from.



Current Printers

Printer	Role today	The catch
Kobra Max	Big prints	Wheel-based motion → visible print artifacts; bed needs frequent re-leveling
Ender 3 V3 SE	Son's 0.4 mm printer	Small, basic, aging

- Both are **single-material** and **open** (no ABS).
- Both are **bedslingers** — the bed moves back and forth, which slows printing and causes vibrations in the print.

The proposal: a Voron 2.4 R2 (300 mm)

open-source

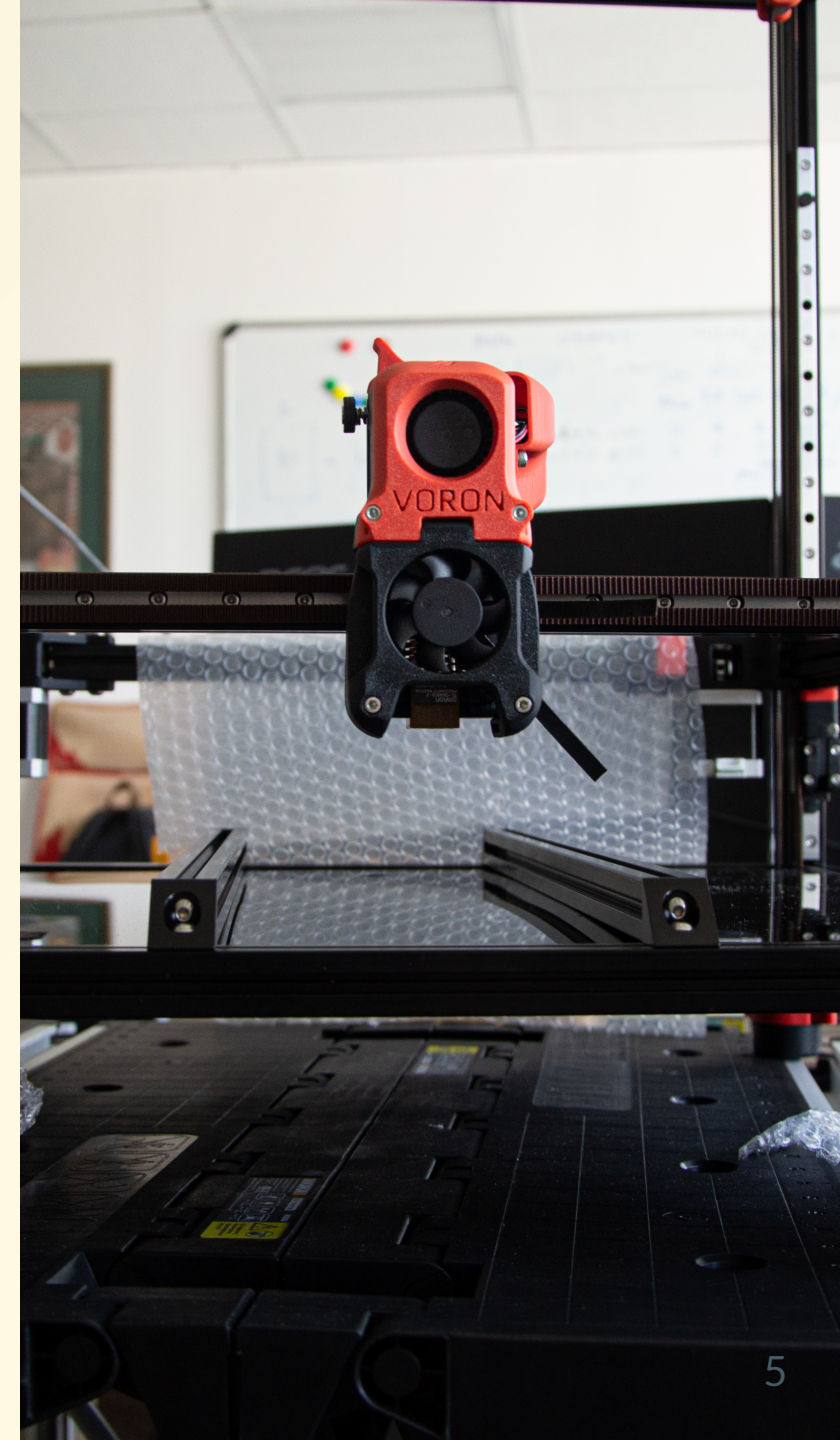
CoreXY

enclosed

multi-material

A self-sourced, fully-enclosed CoreXY printer — the bed stays still, a light gantry flies, and it **auto-levels itself** every print. What goes in it:

Module	Key choices
Toolhead	Revo hotend, Cartographer probe, Galileo 2, filament cutter
Motion & frame	Linear rails, BTT Octopus, Raspberry Pi 5, MeanWell PSU
Bed	Cast aluminum + AC heater + PEI — flat, fixed, stable
Enclosure	Hybrid panels, chamber heater, carbon filter, LED, camera
Multi-material	BoxTurtle auto filament changer — <i>added in a later phase</i>



Why 300 mm (not 250 or 350)?

A **250** would be cheaper — but not by much:

	250 build	300 build
Estimated cost	≈ \$1,773	≈ \$1,845
Difference		only ≈ \$72 (~4%)
Build volume	15.6 L	27 L

- Most parts — electronics, motors, motion, heater, toolhead — **cost the same at any size.**
- Dropping to 250 saves ~4% but **gives up ~42% of the build volume** — a 300 fits **73% more** than a 250.
- A **350** is an option too, but may be harder to calibrate.
- → **300 is the sweet spot.**

What happens to the other printers?

Several ways to go — this is a family decision:

Option	Kobra	Ender	Process	You end up with
A	Keep & specialize	Retire	Build a new Voron from scratch, upgrade the Kobra extruder and y axis	Voron for multi material, fine detail and exotic plastics + Kobra for fast prints and large prints
F ★	Give to son	Retire	Build a new Voron from scratch	Voron for Andreas; Kobra for Joe
B	Strip for parts, then sell	Son keeps	Build Voron parts, get them working on the kobra. Eventually build a Voron frame and moves the parts over	Voron for Andreas; Ender for Joe
G	Sell	Retire	Build a RatRig 400 (~\$2,250+) with a boxturtle	We share one large format, enclosed, multimaterial printer

The budget

Working estimates

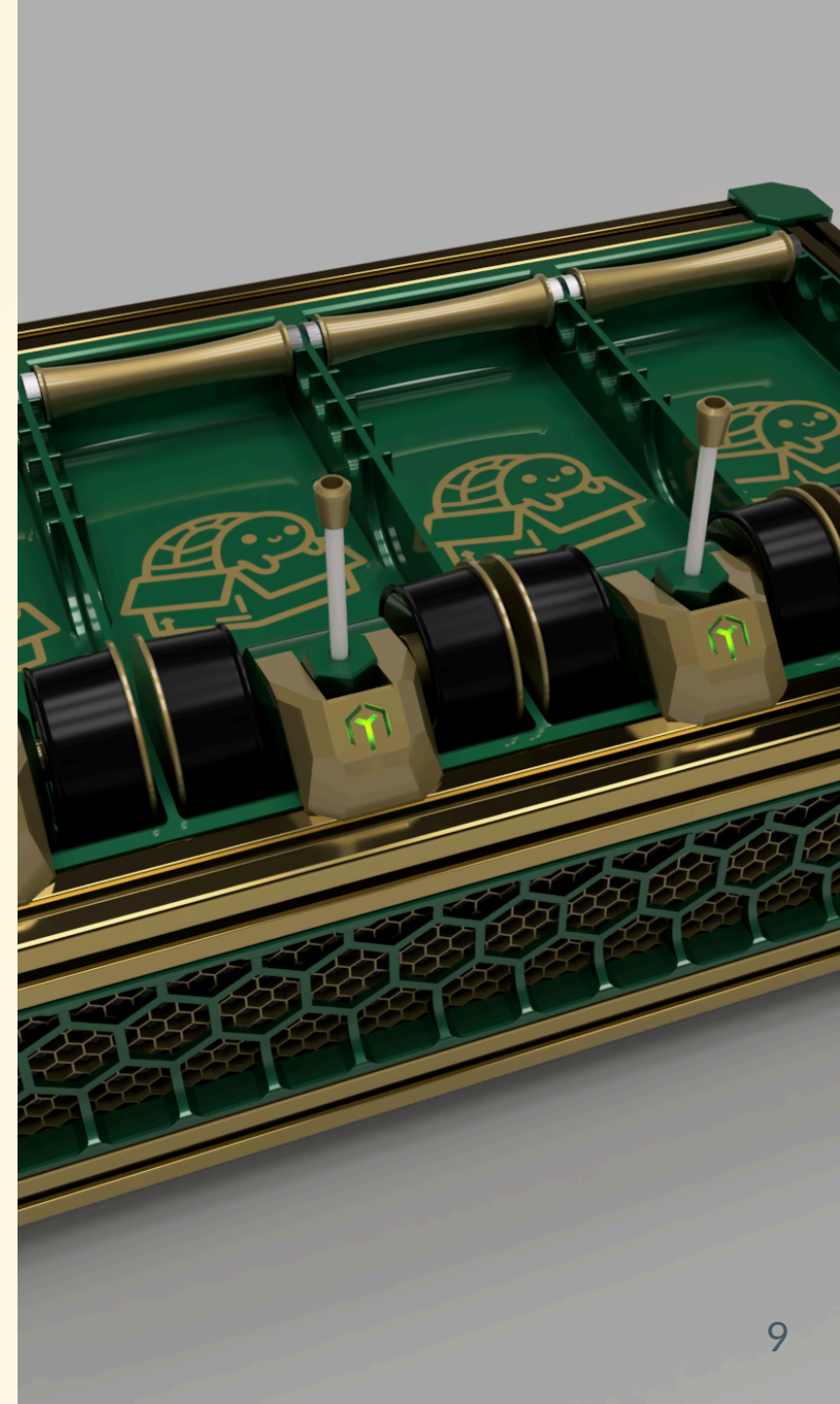
Module	Estimate
Toolhead	≈ \$275
Frame, motion, motors & electronics	≈ \$1,095
Enclosure	≈ \$210
Tools (<i>one-time, reusable</i>)	≈ \$86
Fasteners, wire & connectors	≈ \$181
Core Voron	≈ \$1,846
+ BoxTurtle multi-material (<i>later phase</i>)	+ \$190
All-in	≈ \$2,035

Only the ~\$86 of tools is truly reusable — I keep those forever, for every future print & repair.

Spending it sensibly: phases

The plan is built around **waypoints** — every stage ends with a *working* printer. We can pause at any point.

1. **Print the parts** on our current printers (*≈ filament only*) in petg
2. **Frame + motion + electronics** → a moving, testable machine
3. **Bed + enclosure** → a complete single-material Voron
 - reprint in abs anything that needs to withstand heat
4. **BoxTurtle** → multi-material



The ask

Approval to start building a Voron 2.4 300 — in phases, ~\$1,850–2,050 total, spread over time.

Open questions for us to decide together:

- Which fleet plan (who gets the Kobra)?
- A budget ceiling / per-phase cap?
- Timeline — steady, or "whenever funds allow"?

Detailed line-item pricing lives in the "Printer Upgrades" spreadsheet.

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